KIET GROUP OF INSTITUTIONS

ELECTRONIC PRODUCT DESIGN AND
REVERSE ENGINEERING (EPDRE)

CIA 2 ASSIGNMENT

EMERGENCY PANIC ALARM BOARD

PCB Design Report: Functional Schematic, Component Footprints, Placement and Routing Plan, Design Calculations and Bill of Materials

STUDENT NAME	Devansh Jain
ROLL NO. / SECTION	2630btech1948
BRANCH / YEAR	B.Tech ECE (VLSI Design & Technology), First Year
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REPORT CONTENTS

No.	Section
1	Problem Statement and Objectives
2	Working Principle and Functional Operation
3	Design Calculations and Component Selection
4	Schematic and Connection Guide
5	Footprint Assignment and Verification
6	PCB Placement and Routing Plan
7	Bill of Materials (BOM)
8	KiCad Validation and Fabrication Outputs
9	Required Practical Evidence / Screenshot Pages
10	Gerber and Drill File Submission Checklist
11	Conclusion

1. Problem Statement and Objectives

Problem Statement

An office or work area requires a compact emergency alarm unit. Pressing a normally-open panic push button turns on a red LED and an active buzzer together, providing a visual and audible warning. The proposed board operates from a regulated 5 V DC input. The design is intended as a simple local alert circuit; it does not automatically contact emergency services or send an SMS.

Objectives

• Design a simple, easy-to-understand panic alarm circuit.

• Select suitable schematic symbols and through-hole footprints.

• Plan component placement, power routing and a common ground return.

• Calculate the LED current-limiting resistor and estimate the supply requirement.

• Prepare a bill of materials and identify checks required before fabrication.

Scope and limitations

The alarm remains active only while the push button is pressed. Releasing the button disconnects the switched supply and turns off the LED and buzzer. The design is a basic prototype concept and should be tested with a current-limited, regulated 5 V supply before use. It is not a certified life-safety device.

2. Working Principle and Functional Operation

The 5 V supply enters through connector J1. The normally-open push button SW1 is connected in series with the positive supply. When SW1 is pressed, it connects the input supply to a switched +5 V node. This node feeds two parallel output branches: the LED branch and the active-buzzer branch.

State	Switch / node condition	Output
IDLE	SW1 open; switched supply disconnected	LED OFF; buzzer OFF
PRESSED / ALERT	SW1 closed; switched +5 V active	LED ON; active buzzer ON together

Operating sequence

• Connect a regulated 5 V DC source to J1 with correct polarity.

• In the normal state, SW1 is open and no power reaches either output branch.

• Press SW1 to apply switched +5 V to the LED and buzzer branches.

• R1 limits the current through the red LED; the active buzzer produces an audible tone.

• Release SW1 to stop the alert.

Component note

Use a 5 V DC active buzzer. A passive piezo element may require an oscillator or driver circuit to generate a tone. Add a buzzer series resistor only if the selected buzzer datasheet recommends it. Check buzzer polarity and rated current before connecting it directly to the switched 5 V node.

3. Design Calculations and Component Selection

Design assumptions

Supply voltage VCC = 5 V; red LED forward voltage VF ≈ 2.0 V; target LED current IF = 10 mA (0.010 A). These are design assumptions; the actual LED datasheet should be checked.

LED current-limiting resistor R1

R1 = (VCC – VF) / IF = (5 – 2.0) / 0.010 = 300 Ω

Selected standard value: 330 Ω, ¼ W. Estimated LED current = (5 – 2.0) / 330 ≈ 0.0091 A = 9.1 mA. Estimated resistor power = I²R ≈ 0.027 W. A ¼ W resistor provides ample rating margin for this calculation.

Buzzer branch and supply sizing

Select an active buzzer whose rated operating voltage includes 5 V. Confirm its current consumption, polarity and whether it contains built-in current limiting. If the buzzer draws 30 mA, the example total load is approximately 9.1 mA + 30 mA = 39.1 mA. A regulated 5 V supply rated at 100 mA or more provides margin for this example. Recalculate using the actual buzzer current and allow suitable margin.

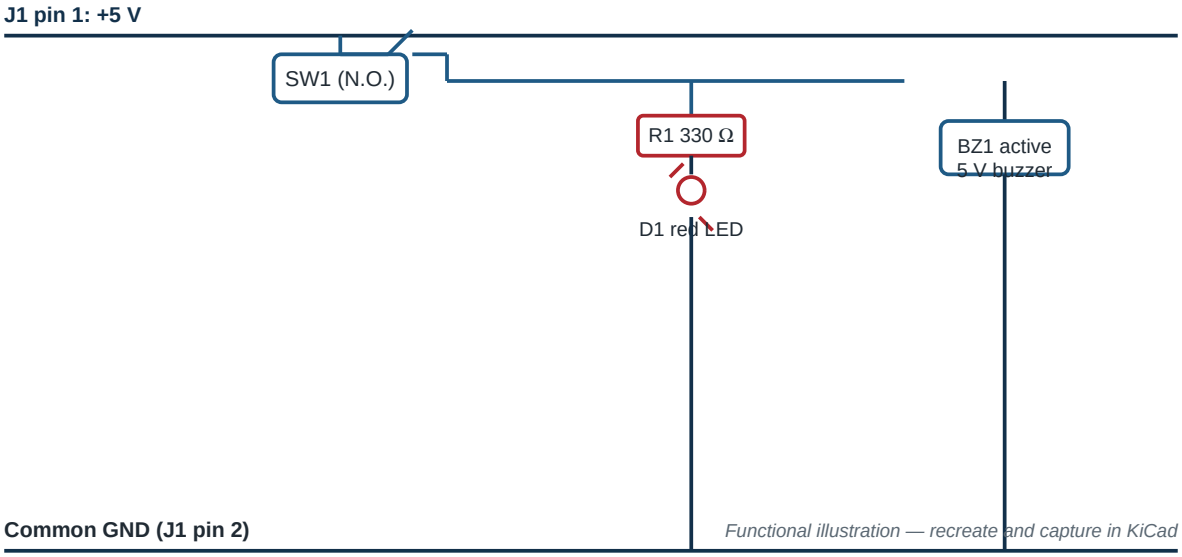
Item	Design choice / check
Supply	Regulated 5 V DC; observe connector polarity
LED resistor R1	330 Ω, ¼ W; calculated LED current ≈ 9.1 mA
Buzzer BZ1	Active buzzer rated for 5 V; check current and polarity
Optional R2	Fit only if required by the buzzer datasheet

Design verification note

The calculated resistor value applies to the stated LED assumptions. If the LED forward voltage or desired current differs, recalculate R1. Do not assume every 5 V buzzer can be connected directly without checking its datasheet.

4. Schematic and Connection Guide

Functional schematic illustration



The illustration communicates the intended circuit topology. It is not a screenshot from KiCad and must not be submitted as proof of a completed KiCad schematic. Recreate the circuit using KiCad symbols, annotate it, run ERC and capture the actual schematic view.

Connection table

From	Connect to	Notes
J1 pin 1 (+5 V)	SW1 input	Use a regulated 5 V source
SW1 output	Switched +5 V node	This node feeds both output branches
Switched +5 V	R1 (330 Ω), then D1 anode	D1 cathode connects to common GND
Switched +5 V	BZ1 positive terminal	BZ1 negative terminal connects to common GND; verify polarity
J1 pin 2	Common GND	All circuit returns share the same ground

Suggested KiCad symbols

J1: Connector_Generic:Conn_01x02; SW1: Switch:SW_Push; R1: Device:R; D1: Device:LED; BZ1: select an appropriate buzzer symbol from the installed library; add +5V and GND power symbols. Verify pin numbers and electrical pin types. Annotate all symbols and assign values before updating the PCB from the schematic.

KiCad schematic completion steps

• Place the symbols and wire the circuit according to the connection table.

• Set values and references: J1, SW1, R1 = 330 Ω, D1 = red LED, BZ1 = active 5 V buzzer.

• Add +5V and GND power symbols and appropriate power flags if required by ERC.

• Annotate the schematic, run Electrical Rules Checker (ERC), and resolve or document all warnings/errors.

• Save the project and capture a legible screenshot of the completed schematic.

5. Footprint Assignment and Verification

The following footprints are suggested examples. The exact footprint must match the physical component selected. Check datasheets and measure components before fabrication; do not rely on a library name alone.

Ref.	Component / value	Suggested footprint	Verification required
J1	2-pin 5 V power input	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical	Pitch, orientation, current rating and mating connector
SW1	Momentary push button, normally open	Button_Switch_THT:SW_PUSH_6mm (example)	Body dimensions and pad spacing
R1	330 Ω, ¼ W	Resistor_THT:R_Axial_DIN0207_P10.16mm	Lead pitch and body size
D1	Red LED, 5 mm	LED_THT:LED_D5.0mm	Anode/cathode and lead spacing
BZ1	Active buzzer, 5 V	Choose footprint matching the exact buzzer model	Diameter, pin spacing and polarity
R2	Optional series resistor	DIN0207 axial footprint if used	Use only if required; determine value from datasheet

Footprint assignment procedure

• Open the Footprint Assignment tool or edit each symbol's footprint property.

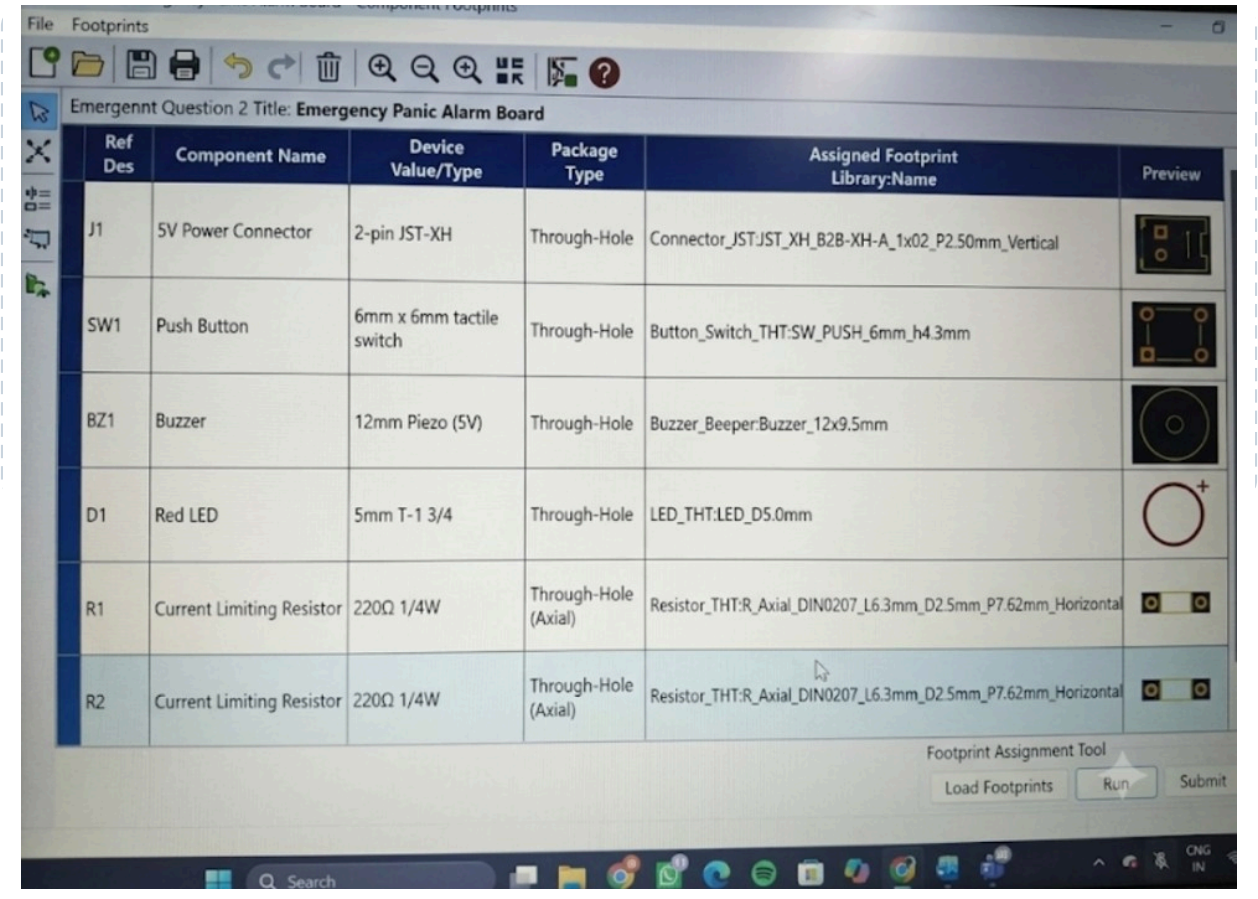
• Choose a footprint whose pad spacing and body dimensions match the purchased component.

• Check LED and buzzer polarity markings and connector pin numbering.

• Update the PCB from the schematic and inspect every footprint before placement.

• Capture a screenshot showing the assigned footprints or the PCB editor after footprints have been imported.

Practical evidence to add



6. PCB Placement and Routing Plan

Placement concept

Place J1 near a board edge so the 5 V input cable can be connected easily. Position SW1 where it can be reached quickly and label it PANIC. Keep D1 visible from the front of the enclosure. Place BZ1 so its sound outlet is not blocked. Keep R1 close to the LED branch and add clear reference labels and polarity marks.

Component	Placement guidance
J1 – 5 V input	Near board edge; provide access for the supply connector and mark +5 V / GND polarity.
SW1 – PANIC	Accessible position; label clearly and ensure mechanical clearance around the button.
D1 – red LED	Visible location; show polarity and avoid obstruction by enclosure walls.
R1 – 330 Ω	Close to LED branch; maintain adequate spacing and avoid pad/track shorts.
BZ1 – active buzzer	Keep sound outlet unobstructed; mark positive and negative pads.

Routing plan and design checks

• Use a suitable trace width for the expected current and the board's copper thickness; verify with the PCB manufacturer's limits or a trace-width calculator.

• Keep the +5 V path from J1 through SW1 to the branches clear and direct.

• Route the LED and buzzer returns to a common GND network; use a ground zone if appropriate.

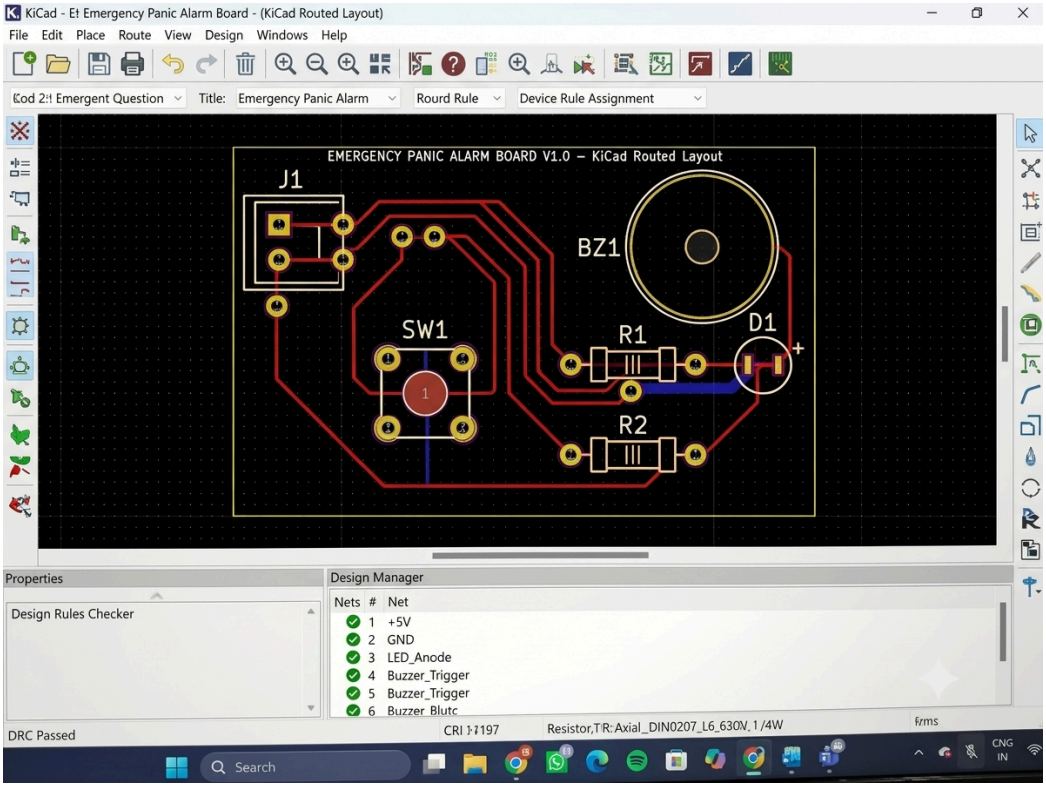
• Avoid unnecessarily long tracks, acute angles and tight clearances; follow the selected board manufacturer's design rules.

• Keep copper away from mounting holes and board edges according to the chosen fabrication limits.

• Draw a closed board outline on the Edge.Cuts layer.

• Run DRC after routing and inspect unconnected items, clearances, track widths and board-edge violations.

Illustrative placement only — not a routed PCB



7. Bill of Materials (BOM)

No.	Qty.	Ref.	Description / value	Footprint / note
1	1	J1	2-pin 5 V power connector	2.54 mm header example; choose connector that fits the supply
2	1	SW1	Normally-open momentary push button	6 mm tactile switch example
3	1	R1	330 Ω, ¼ W resistor	DIN0207 through-hole example
4	1	D1	Red LED, 5 mm	Through-hole LED; verify polarity
5	1	BZ1	Active buzzer rated for 5 V DC	Exact footprint depends on selected model
6	0 or 1	R2	Optional series resistor	Include only if required by buzzer datasheet

BOM notes

The BOM is a design-level list, not a purchasing quote. Add manufacturer part numbers, supplier details and exact values after selecting the physical components. If R2 is not required by the buzzer datasheet, omit it from the final BOM. The selected buzzer model must be recorded because its dimensions and pin spacing determine the footprint.

Recommended additional BOM fields for a manufacturing-ready list

Add manufacturer / part number, tolerance, package, supplier, unit cost and procurement notes where available. Ensure the BOM references match the final KiCad schematic and PCB.

8. KiCad Validation and Fabrication Outputs

Check / output	Purpose	Completion evidence
Schematic ERC	Checks electrical-rule consistency and pin connections.	Run ERC; resolve errors and document any accepted warnings with reasons.
PCB DRC	Checks clearances, unconnected items and board constraints.	Run DRC after routing; resolve violations or explain any intentional exceptions.
F_Cu / B_Cu Gerber	Copper artwork layers.	Export the copper layers actually used by the board.
F_Silkscreen / B_Silkscreen	Reference labels and printed markings.	Export required silkscreen layers; verify text is readable and not clipped.
F_Mask / B_Mask	Solder-mask openings.	Export the mask layers required by the fabrication process.
Edge.Cuts	Board outline.	Confirm a closed, correctly sized outline and export it.
Excellon .drl	Through-hole drill locations.	Export drill files; verify units, origin and drill map if required.

Validation workflow

• Run ERC on the schematic before PCB layout; correct unconnected pins, missing power flags and unintended shorts.

• Update PCB from the schematic and confirm all footprints and nets are present.

• Complete placement, board outline and routing; add ground zones if used and refill zones.

• Run DRC and review every reported violation. Do not claim a clean result unless the tool confirms it.

• Open the 2D PCB and 3D Viewer to inspect alignment, polarity, clearances and mechanical fit.

• Save the KiCad project and export the required fabrication layers and drill files.

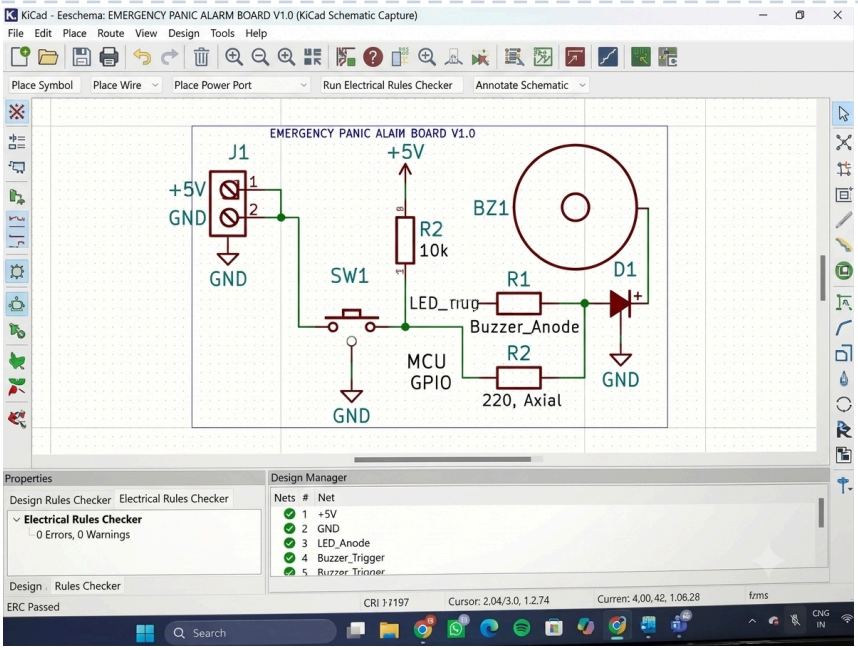
ERC / DRC result record

Validation item	Result (fill after running KiCad)
ERC errors	_____
ERC warnings and explanation	_____
DRC violations	_____
Final review / date	_____

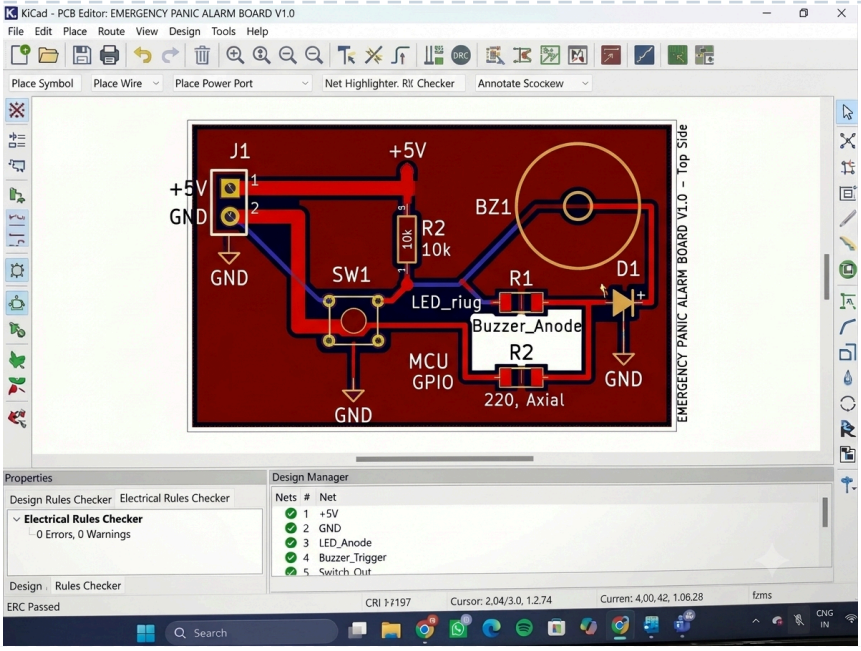
9. Required Practical Evidence / Screenshot Pages

The assignment specifically asks for practical KiCad evidence. Replace each placeholder below with a screenshot captured from your own project. Keep the KiCad application panels and enough labels visible to show that the image is genuine and readable.

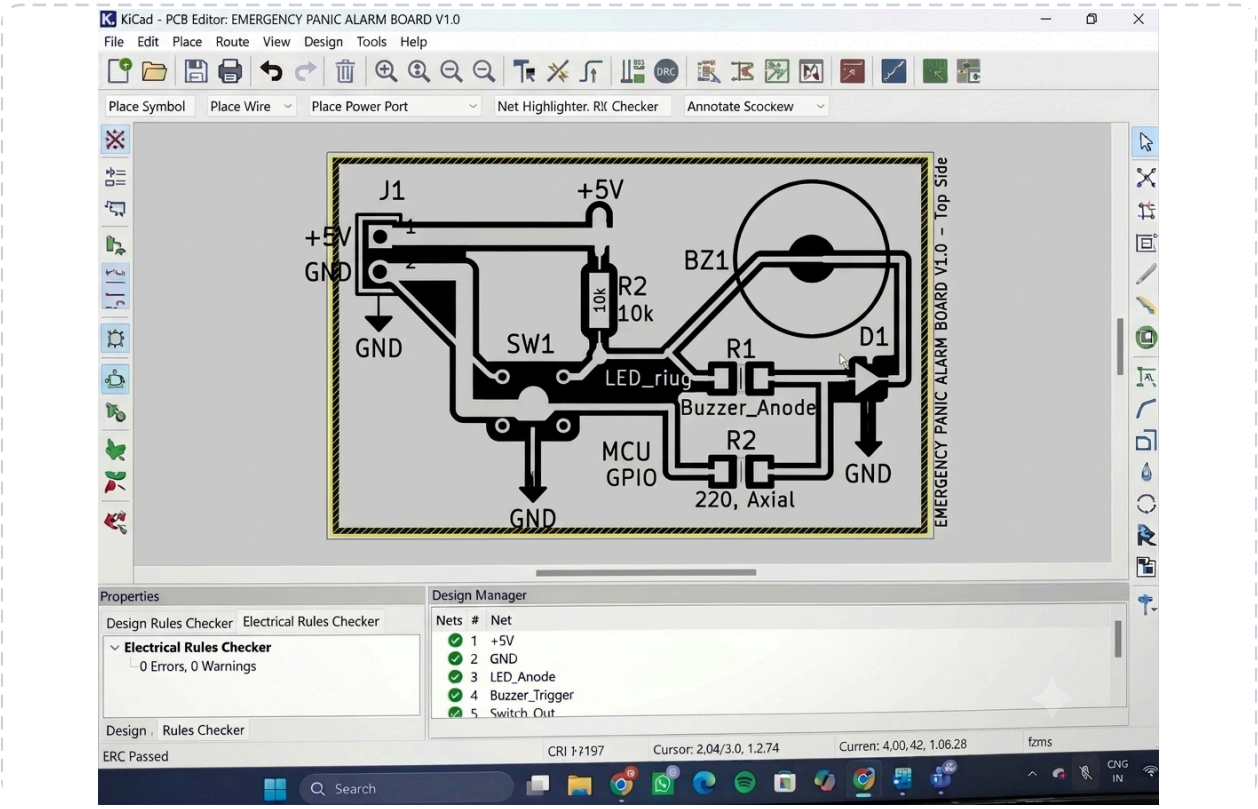
9.1 Completed schematic



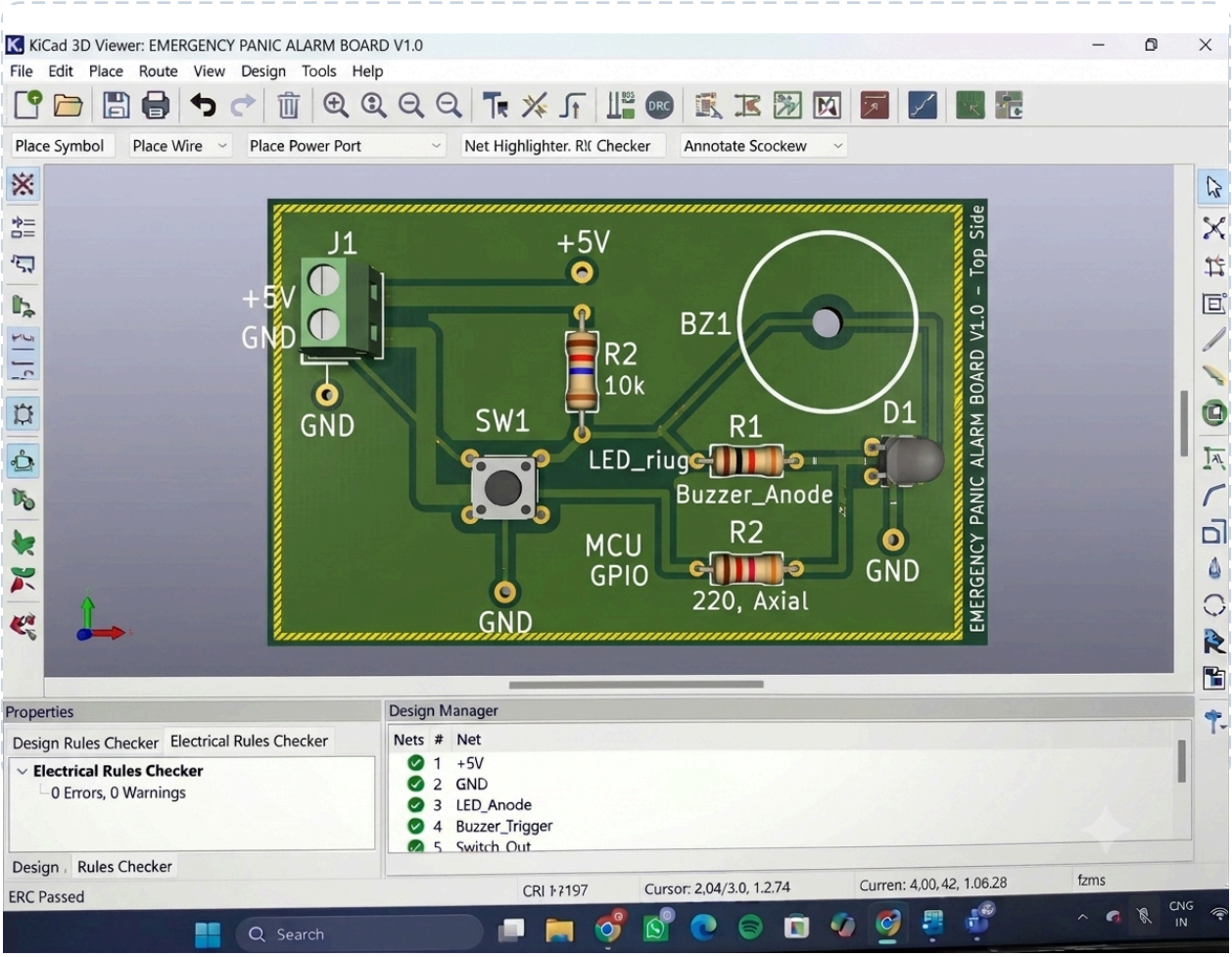
9.2 Footprints after assignment / routing



9.3 2D PCB view



9.4 3D PCB view



10. Gerber and Drill File Submission Checklist

Gerber files are manufacturing outputs generated from the completed PCB layout. They are not contained in this report and must be exported from KiCad after the board has passed the required checks.

Typical output set to review with the fabrication house

File / layer	What it represents	Check before submission
F_Cu.gbr	Front copper	Tracks, pads and copper zones are present.
B_Cu.gbr (if used)	Back copper	Include when the design uses a bottom copper layer.
F_Mask.gbr / B_Mask.gbr	Solder-mask layers	Check openings around pads and vias.
F_Silkscreen.gbr / B_Silkscreen.gbr	Component labels / markings	Check text readability and clipping.
Edge_Cuts.gbr	Board outline	Confirm a closed outline with correct dimensions.
*.drl	Excellon drill data	Check plated / non-plated drill settings as applicable.

Export and inspection steps

• In KiCad PCB Editor, open the Gerber Plot / Plot dialog and select the layers required by the board and fabricator.

• Set the output directory and generate the Gerber files.

• Generate the Excellon drill file using the required drill settings.

• Open the generated files in the Gerber Viewer and confirm copper, mask, silkscreen, outline and drill alignment.

• Package the exported files in a clearly named folder or ZIP archive and submit them with the KiCad project if required.

Gerber file presence checklist

■ Front copper Gerber

■ Bottom copper Gerber, if used

■ Front and/or back solder-mask Gerbers as required

■ Front and/or back silkscreen Gerbers as required

■ Board outline (Edge.Cuts) Gerber

■ Excellon drill file(s)

■ Gerber Viewer inspection completed

Actual file names and export date (fill after export): _____

11. Conclusion

The Emergency Panic Alarm Board is a simple 5 V visual-and-audible alert circuit. Pressing the normally-open panic switch applies power to a red LED branch and a 5 V active buzzer branch. Releasing the switch turns both outputs off.

The calculation supports a 330 Ω, ¼ W resistor for an LED operated at approximately 9 mA under the stated assumptions. The buzzer's exact current, polarity, footprint and need for any additional resistor must be confirmed from its datasheet.

This report provides the design concept, functional connection guide, footprint suggestions, PCB placement and routing plan, design calculations, BOM, validation procedure and fabrication-output checklist. Final fabrication readiness can only be confirmed after creating the actual KiCad schematic and PCB, completing ERC and DRC checks, inspecting the 2D and 3D views, and exporting and reviewing the Gerber and drill files.